

Assignment 3

Introduction to IP Addressing Theory

Task (1):

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Example: 192.168.1.10	Example: 2001:db8::10
Provides about 4.3 billion possible addresses.	Provides a vastly larger address space (about 3.4×10^{38} addresses).
Addresses are written in decimal , separated by dots.	Addresses are written in hexadecimal , separated by colons.
Uses ARP to map IP addresses to MAC addresses.	Uses Neighbor Discovery Protocol (NDP) instead of ARP.
NAT is commonly used because public IPv4 addresses are limited.	NAT is generally not required , because the address space is extremely large.
Broadcast is supported.	No broadcast ; IPv6 uses multicast and anycast instead.
Security support (IPsec) is available but not mandatory.	IPsec support is built into the IPv6 specification , though its use is not universally mandatory in every deployment.

Task (2):

Device	Static IP Address	Subnet Mask	Default Gateway
 Laptop	192.168.10.10	255.255.255.0	192.168.10.1
 Phone	192.168.10.11	255.255.255.0	192.168.10.1
 Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
 Printer	192.168.10.13	255.255.255.0	192.168.10.1

Task (3):

A /27 mask is 255.255.255.224, leaving 5 host bits.

- Hosts per subnet: $2^5 - 2 = 30$ usable hosts
- Number of subnets (carved out of the original /24): $2^3 = 8$ subnets

Subnet	Network Address	Broadcast Address	Usable Host IPs
1	192.168.1.0/27	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32/27	192.168.1.63	192.168.1.33 – 192.168.1.62
3	192.168.1.64/27	192.168.1.95	192.168.1.65 – 192.168.1.94

Task (4):

10.0.5.17 /24 with mask 255.255.255.0

Network address: 10.0.5.0

Broadcast address: 10.0.5.255

Valid host range: 10.0.5.1 – 10.0.5.254